

9. (2.25 points)

An insurer has written a book of workers compensation business for 20 years with a per-occurrence deductible of \$250,000.

Given the following insurer and industry information:

Hazard Group	Insurer Premium Distribution	Industry Unlimited Expected Loss Ratio	Industry Excess Ratio at \$250,000
1	50%	45%	0.05
2	30%	50%	0.10
3	10%	50%	0.20
4	10%	60%	0.40

Given the following information for the insurer:

- All 2014 claims have been reported by December 31, 2015
- 2014 earned premium is \$50,000,000
- 2014 reported insurer-retained losses as of December 31, 2015 are \$2,500,000

Unlimited severity at ultimate	\$5,000
Unlimited severity at 24 months	3,200
Deductible severity at ultimate	2,000
Deductible severity at 24 months	1,500

a. (0.5 point)

Estimate the 2014 ultimate insurer-retained losses, incorporating the given industry information.

b. (1.25 points)

Estimate the 2014 ultimate insurer-retained losses using only the insurer's information.

c. (0.5 point)

Provide one reason for and one reason against selecting the estimate calculated in part a. above as opposed to the estimate calculated in part b. above.

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EXAM 7 SPRING 2016 SAMPLE ANSWERS AND EXAMINER'S REPORT

QUESTION 9							
TOTAL POINT VALUE: 2.25	LEARNING OBJECTIVE: A4: Estimate unpaid claims for various layers of claims.						
SAMPLE ANSWERS							
Part a: 0.5 point							
<i>Sample Answer 1</i> Take weighted expected excess loss from industry $0.5 \times 0.45 \times 0.05 + 0.3 \times 0.5 \times 0.1 + \dots$ $= 0.06025 \times 50,000 = 3,012,500$ <i>Sample Answer 2</i> $\text{Ult} = 50,000,000 \times (50\% \times 45\% \times 0.05 + 30\% \times 50\% \times 0.1 + 10\% \times 50\% \times 0.2 + 50\% \times 60\% \times 0.6) = 3,012,500$							
Part b: 1.25 points							
<i>Sample Answer 1</i> $R^L = 2000/5000 = .4$ $R_t^L = 1500/3200 = .46875$ $\text{Unlimited LDF} = 5000/3200 = 1.5625$ $\text{XSLDF} = 1.5625 \times (1 - .4)/(1 - .46875) = 1.7647$ $\text{Ult XS Loss} = 2.5\text{M} \times 1.7647 = 4,411,750$ <i>Sample Answer 2</i> As all reported claims have been reported as of 24 months, all LDFS are based on excess severity development: $(5000 - 2000)/(3200 - 1500) = 1.765$ $1.765 \times 2,500,000 = 4,412,500$ <i>Sample Answer 3</i> $3200 - 1500 = 1700 \text{ (excess severity at 24 months)}$ $2,500,000/1700 = 1471 \text{ (\# of excess claims)}$ $1471 \times (5000 - 2000) = 4,413,000$ <i>Sample Answer 4</i> <table border="0"> <tr> <td>Unlimited LDF</td> <td>1.563 = 5000/3200</td> </tr> <tr> <td>Limited LDF</td> <td>1.333 = 2000/1500</td> </tr> <tr> <td>limited to unlimited ratio @24 months</td> <td>0.469 = 1500/3200</td> </tr> </table> Formula : Unlimited LDF = Limited LDF × Limited to Unlimited Ratio + Excess LDF × (1- Limited to Unlimited Ratio) $1.563 = 1.333 \times .469 + \text{Excess LDF} \times (1-.469)$		Unlimited LDF	1.563 = 5000/3200	Limited LDF	1.333 = 2000/1500	limited to unlimited ratio @24 months	0.469 = 1500/3200
Unlimited LDF	1.563 = 5000/3200						
Limited LDF	1.333 = 2000/1500						
limited to unlimited ratio @24 months	0.469 = 1500/3200						

EXAM 7 SPRING 2016 SAMPLE ANSWERS AND EXAMINER'S REPORT

Solve for Excess LDF = 1.765 Therefore $1.765 \times \\$2,500,000 = \\$4,411,765$
Part c: 0.5 point
<i>Sample Answer 1</i> For: Loss ratio method in a. can provide more stable result while b. can be very volatile especially when losses reported are extreme. Against: a. does not take into account actual loss experience. <i>Sample Answer 2</i> For: much more stable at early maturities where xs losses are thin. Against: ignores actual emergence of losses. <i>Sample Answer 3</i> • Based on insurer's data – since the insurer has been in business a long time, their own data better reflects the characteristics of their book of business, which may be different than the industry. • Using industry data – the industry information is more credible and less subject to distortions from individual company experience.
EXAMINER'S REPORT
Part a Candidates were expected to use the hazard group information with 2014 earned premium to derive an estimate of ultimate losses for 2014. The candidates performed well on part a. A common error was to think that losses under the deductible are insurer retained losses. Another common error was to use the premium $\times$ industry unlimited expected loss ratio correctly, but in a silo computing the industry excess ratio using the premiums as the only weights as opposed to using the derived losses as weights.
Part b Candidates were expected to use the severity information with the 2014 reported losses to derive a separate estimate of 2014 ultimate losses. Candidates performed reasonably well, generally receiving full credit for getting the correct answer via a total of 4 different approaches, as outlined above. A common error from the suggested answer is making the assumption that since all claims are reported within 24 months, the unlimited LDF is then 1.00. This ignores development on known claims which makes this assumption wrong. This fails to consider the severity development (\$5k/\$3.2K).
Part c For each method above, candidates were expected to provide a reason to rely more on it rather than the other one. Candidates fared very well on this part, though legibility was sometimes a problem, leading to no credit awarded.